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241401113

OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP – 05: SCHEDULING ALGORITHMS (First Come First Serve)

```
Activities    Terminal                    Mar 25  00:07
                     user123@localhost:~/scheduling
[user123@localhost ~]$ ls
abc.sh  Desktop    Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh   SFA  Templates  while.sh
ab.sh   Documents  Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  sh   Videos
[user123@localhost ~]$ mkdir scheduling
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ vi fcfs.c
[user123@localhost scheduling]$ cat fcfs.c
#include <stdio.h>

int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d",&n);

        int at[n], bt[n], ct[n], tat[n], wt[n];

        for(int i=0; i<n; i++){
                printf("Enter Arrival time and Burst Time for P%d: ", i+1);
                scanf("%d %d:",&at[i],&bt[i]);
        }

        float total_tat = 0, total_wt = 0;

        ct[0] = at[0] + bt[0];

        for(int i=0; i<n; i++){
                if(ct[i-1]<at[i]){
                        ct[i] = at[i] + bt[i];
                } else {
                        ct[i] = ct[i-1] + bt[i];
                }
        }

        for(int i=0; i<n; i++){
                tat[i] = ct[i] - at[i];
                wt[i] = tat[i] - bt[i];

                total_tat += tat[i];
                total_wt += wt[i];
        }

        printf("\nAverage Turnaround Time: %.2f", total_tat/n);
        printf("\nAverage Waiting Time: %.2f", total_wt/n);

        return 0;
}
[user123@localhost scheduling]$ gcc fcfs.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes: 4
Enter Arrival time and Burst Time for P1: 0 2
Enter Arrival time and Burst Time for P2: 1 2
Enter Arrival time and Burst Time for P3: 5 3
Enter Arrival time and Burst Time for P4: 6 4

Average Turnaround Time: 3.50
Average Waiting Time: 0.75[user123@localhost scheduling]$
```

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OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP – 05: SCHEDULING ALGORITHMS (Shortest Job First)

```
Activities    Terminal                Mar 25 00:44
                    user123@localhost:~/scheduling
[user123@localhost ~]$ ls
abc.sh  Desktop    Downloads  ifelse.sh  Music  Pictures  Public   REC       sample.sh   scheduling  sh         Videos
ab.sh   Documents  Folder     loops.sh   pc     posneg.sh qns.txt  sample.c  sample.txt  SFA         Templates  while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ vi sjf.c
[user123@localhost scheduling]$ cat sjf.c
#include <stdio.h>

int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d",&n);

        int at[n], bt[n], ct[n], tat[n], wt[n];
        int done[n];

        for(int i=0; i<n; i++){
                printf("Enter Arrival Time and Burst Time for P%d: ",i+1);
                scanf("%d %d",&at[i], &bt[i]);
                done[i] = 0;
        }

        int completed = 0;
        int completionTime = 0;
        float avgtat = 0, avgwt = 0;

        while(completed < n){
                int idx = -1;
                int minBT = 9999;

                for(int i=0; i<n; i++){
                        if(at[i]<=completionTime && done[i] == 0){
                                if(bt[i]<minBT){
                                        minBT = bt[i];
                                        idx = i;
                                }
                        }
                }

                if(idx != -1){
                        if(completionTime<at[idx]){
                                completionTime = at[idx];
                        }
                        completionTime += bt[idx];

                        tat[idx] = completionTime - at[idx];
                        wt[idx] = tat[idx] - bt[idx];

                        done[idx] = 1;
                        completed++;
                } else {
                        completionTime++;
                }
        }

        for(int i=0; i<n; i++){
                avgtat += tat[i];
                avgwt += wt[i];
        }

        printf("\nAverage Turnaround Time: %.2f\n",avgtat/n);
        printf("Average Waiting Time: %.2f\n",avgwt/n);

        return 0;
}
[user123@localhost scheduling]$
```

```
[user123@localhost scheduling]$ gcc sjf.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 1 3
Enter Arrival Time and Burst Time for P2: 2 4
Enter Arrival Time and Burst Time for P3: 1 2
Enter Arrival Time and Burst Time for P4: 4 4

Average Turnaround Time: 6.25
Average Waiting Time: 3.00
[user123@localhost scheduling]$ [redacted]
```

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OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP – 5: SCHEDULING ALGORITHMS (Priority Scheduling)

```
user123@localhost:~/scheduling
[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures  Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  sjf.c
[user123@localhost scheduling]$ vi priority.c
[user123@localhost scheduling]$
```

S

```c
user123@localhost:~/scheduling — /usr/bin/vim priority.c
#include <stdio.h>
int main()
{
    int n, completed = 0, time = 0;
    printf("Enter number of processes: \n");
    scanf("%d", &n);
    int at[n], bt[n], priority[n], done[n];
    int ct[n], tat[n], wt[n];
    float avgtat = 0, avgwt = 0;
    // Input
    for (int i = 0; i < n; i++)
    {
        printf("Enter AT, BT and Priority for P%d: \n", i + 1);
        scanf("%d %d %d", &at[i], &bt[i], &priority[i]);
        done[i] = 0;
    }
    // Scheduling Logic
    while (completed < n)
    {
        int idx = -1;
        int highPriority = 9999;
        for (int i = 0; i < n; i++)
        {
            if (at[i] <= time && done[i] == 0)
            {
                if (priority[i] < highPriority)
                {
                    highPriority = priority[i];
                    idx = i;
                }
            }
        }
        if (idx != -1)
        {
            time += bt[idx];
            ct[idx] = time;
            tat[idx] = ct[idx] - at[idx];
            wt[idx] = tat[idx] - bt[idx];
            done[idx] = 1;
            completed++;
        }
        else
        {
            time++; // CPU Idle
        }
    }
    // Separate loop for average calculation
    for (int i = 0; i < n; i++)
    {
        avgtat += tat[i];
        avgwt += wt[i];
    }
    printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);
    printf("Average Waiting Time: %.2f\n", avgwt / n);
    return 0;
}
```

user123@localhost:~/scheduling

```
[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  sjf.c
[user123@localhost scheduling]$ vi priority.c
[user123@localhost scheduling]$ gcc priority.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes:
4
Enter AT, BT and Priority for P1:
0 5 10
Enter AT, BT and Priority for P2:
1 4 20
Enter AT, BT and Priority for P3:
2 2 30
Enter AT, BT and Priority for P4:
4 1 40

Average Turnaround Time: 7.50
Average Waiting Time: 4.50
```

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OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP – 5: SCHEDULING ALGORITHMS (Round Robin Scheduling)

user123@localhost:~/scheduling

```
[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures    Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh   qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  priority.c  sjf.c
[user123@localhost scheduling]$ vi rr.c
```

user123@localhost:~/scheduling — /usr/bin/vim rr.c

```c
#include <stdio.h>
int main(){
        int n, tq, completed = 0, completionTime = 0;

        printf("Enter number of processes: \n");
        scanf("%d", &n);

        printf("Enter Time Quantum: \n");
        scanf("%d", &tq);

        int at[n], bt[n], rem[n], tat[n], wt[n];
        float avgtat = 0, avgwt = 0;
        // Input
        for (int i = 0; i < n; i++)
        {
                printf("Enter AT and BT for P%d: \n", i + 1);
                scanf("%d %d", &at[i], &bt[i]);
                rem[i] = bt[i]; // Remaining time = Burst time
        }
        // Round Robin Logic
        while (completed < n)
        {
                int idle = 1;
                for (int i = 0; i < n; i++)
                {
                        if (at[i] <= completionTime && rem[i] > 0) //
                        {
                                idle = 0; //
                                if (rem[i] > tq) //3,5 /1>2
                                {
                                        completionTime += tq; //1+2 ==3
                                        rem[i] -= tq; //5-2=3
                                } else {
                                        completionTime += rem[i]; //
                                        tat[i] = completionTime - at[i]; // Turnaround Time
                                        wt[i] = tat[i] - bt[i]; // Waiting Time
                                        rem[i] = 0;
                                        completed++;
                                }
                        }

                }
                if (idle){
                        completionTime++; // If no process arrived yet//1
                }
        }

        // Calculate averages
        for (int i = 0; i < n; i++)
        {
                avgtat += tat[i];
                avgwt += wt[i];
        }

        printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);

        printf("Average Waiting Time: %.2f\n", avgwt / n);

        return 0;
}
```

user123@localhost:~/scheduling

```
[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd scheduling
[user123@localhost scheduling]$ ls
a.out  fcfs.c  priority.c  sjf.c
[user123@localhost scheduling]$ vi rr.c
[user123@localhost scheduling]$ gcc rr.c
[user123@localhost scheduling]$ ./a.out
Enter number of processes:
4
Enter Time Quantum:
2
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 4
Enter AT and BT for P3:
2 2
Enter AT and BT for P4:
4 1

Average Turnaround Time: 7.25
Average Waiting Time: 4.25
```